

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 3 – Comparative Analysis

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 3 – Comparative Analysis
Weightage:	20% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	B.Tech AI &DS	Date:	08-08-2026

Code of Conduct

I ___N.Kishan Kumar-192424404_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- Read each scenario carefully before answering.
- Identify the NLP techniques being compared in the question.
- Analyze each approach based on its working principles, advantages, limitations, and applicability.
- Compare the alternatives using technical reasoning rather than listing definitions.
- Support your analysis with examples from the given scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze sentence representations, compare structural representations of language, and evaluate how different parsing approaches capture meaning and relationships among words.	Q1
Syntax-Driven Semantic Analysis	Analyze parsing strategies for incomplete and ambiguous inputs, evaluate parsing efficiency, and determine suitable methods for real-time language understanding.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze ambiguity in natural language sentences, evaluate ambiguity resolution techniques, and determine the most effective approach for selecting intended meanings.	Q3
Representing Linguistically Relevant Concepts	Analyze grammatical constraints, agreement features, argument structures, and linguistic representations required for accurate language processing.	Q4
Information Retrieval & Syntax-Driven Semantic Analysis	Analyze large-scale parsing strategies, evaluate performance trade-offs, and justify efficient approaches for processing massive linguistic datasets.	Q5

Q1. CFG Trees vs Dependency Parsing

A language processing system needs to represent sentence structure for grammar checking and syntactic analysis. Two commonly used approaches are Context-Free Grammar (CFG) trees and dependency parsing. CFG represents a sentence using hierarchical constituents, while dependency parsing represents relationships directly between words.

CFG Trees

CFG represents a sentence using phrases such as noun phrases and verb phrases. It shows how smaller grammatical units combine to form larger structures. For example, a sentence can be divided into a subject, verb phrase, and object.

Advantages:

- Clearly represents hierarchical sentence structure.
- Useful for grammar checking and syntactic analysis.
- Provides detailed phrase-level information.
- Suitable for rule-based language processing.

Limitations:

- Large grammar rules can become complex.
- Long sentences may require more parsing effort.
- Direct relationships between individual words are less explicit.

Dependency Parsing

Dependency parsing represents a sentence through relationships between individual words. One word acts as the head and other words depend on it. For example, in “The student reads a book,” “reads” can be the main head, with “student” and “book” connected to it.

Advantages:

- Clearly represents relationships between words.
- Useful for semantic and information extraction tasks.
- Handles long-distance word relationships effectively.
- Provides a compact representation.

Comparison and Conclusion

CFG is better for understanding hierarchical phrase structures, whereas dependency parsing is better for identifying direct relationships between words. For applications where relationships among words are important, dependency parsing is more effective. Therefore, dependency parsing is preferable for capturing word-to-word relationships, while CFG remains useful for detailed grammatical structure.

Q2. Top-Down Parsing vs Earley Parsing

A real-time NLP application must process incomplete and ambiguous user input efficiently. The existing system uses top-down parsing, while Earley parsing is considered as an alternative.

Top-Down Parsing

Top-down parsing begins with the start symbol of the grammar and attempts to generate the input sentence. It predicts the required grammar structures before matching them with the input.

Advantages:

- Simple and easy to understand.
- Suitable for simple and predictable grammars.
- Easy to implement for restricted applications.

Limitations:

- May require extensive backtracking.
- Ambiguous sentences can increase processing time.
- Incomplete input is difficult to process.
- Not always efficient for complex grammars.

Earley Parsing

Earley parsing processes the input from left to right while maintaining possible partial parse structures. It can handle ambiguous and incomplete sentences and does not require a grammar to be converted into a restricted form.

Advantages:

- Handles ambiguous sentences effectively.
- Supports incomplete and partial inputs.
- Reduces unnecessary backtracking.
- Suitable for general CFG-based parsing.
- Useful for dynamic language applications.

Comparison and Conclusion

Top-down parsing is suitable for simple and predictable inputs but becomes inefficient when ambiguity and incomplete sentences are present. Earley parsing is more flexible and can maintain multiple possible interpretations during processing. Therefore, Earley parsing is more suitable for real-time applications where users may provide incomplete or ambiguous input.

Q3. CFG, PCFG and Neural Parsing

An NLP system must resolve ambiguity in the sentence “She saw the man with a telescope.” The possible interpretations are that she used a telescope to see the man, or the man had the telescope. CFG, PCFG, and neural parsing provide different approaches for handling this ambiguity.

CFG

CFG represents the possible syntactic structures according to predefined grammar rules. It can generate more than one valid parse for an ambiguous sentence.

Advantages:

- Simple rule-based representation.
- Easy to interpret.
- Useful for controlled language.

Limitation:

- Cannot decide the intended meaning when multiple valid structures exist without additional information.

PCFG

PCFG extends CFG by assigning probabilities to grammar rules. When multiple parse trees are generated, their probabilities can be compared and the most probable structure can be selected.

Advantages:

- Handles ambiguity better than basic CFG.

- Uses statistical information for selecting parses.
- Useful when training data is available.

Limitation:

- Depends on the quality of probability estimates and training data.

Neural Parsing

Neural parsing uses machine learning models to learn patterns from large amounts of language data. It can use contextual information to determine the most likely interpretation.

Advantages:

- Handles complex language patterns.
- Uses contextual information.
- Performs well on real-world language.
- Can learn representations automatically.

Limitations:

- Requires large training datasets.
- Requires higher computational resources.
- The reasoning behind some predictions may be difficult to explain.

Comparison and Conclusion

CFG is useful for basic grammatical analysis but has limited ambiguity resolution. PCFG improves CFG by using probabilities. Neural parsing can consider broader context and complex language patterns. For real-world NLP applications, neural parsing is generally the most effective when sufficient training data and computational resources are available.

Q4. Feature Structures vs Subcategorization Frames

A grammar system needs to correctly handle subject–verb agreement and verb argument structures. Feature structures and subcategorization frames provide different methods for representing linguistic information.

Feature Structures

Feature structures represent grammatical properties using features and values. Common features include number, gender, person, tense, and case.

Example:

Noun → Number: Singular

Verb → Number: Singular

This allows the system to check whether the subject and verb agree.

Advantages:

- Handles grammatical agreement effectively.
- Represents multiple linguistic features.
- Helps prevent grammatically incorrect combinations.
- Useful for detailed linguistic representation.

Subcategorization Frames

Subcategorization frames describe the arguments required by a verb. Different verbs require different complements.

For example:

give → Subject + Object + Indirect Object

recommend → Subject + Action

These frames help the system understand the relationship between verbs and their required arguments.

Advantages:

- Represents verb argument structures.
- Helps identify relationships between actions and entities.
- Useful for semantic interpretation.
- Supports accurate sentence analysis.

Comparison and Conclusion

Feature structures are more suitable for enforcing grammatical constraints such as subject–verb agreement. Subcategorization frames are more suitable for representing the arguments required by verbs. Therefore, a combined approach provides better language processing because it can handle both grammatical agreement and verb argument relationships.

Q5. Transition-Based Parsing vs Graph-Based Parsing

A large-scale NLP system needs to perform fast and accurate dependency parsing on large datasets. The two approaches being considered are transition-based parsing and graph-based parsing.

Transition-Based Parsing

Transition-based parsing constructs a dependency tree through a sequence of parsing actions. At each step, the parser makes a decision about which transition should be performed.

Advantages:

- Fast processing.
- Requires relatively less computation.
- Suitable for real-time applications.
- Efficient for large datasets.

Limitations:

- Early incorrect decisions can affect later results.
- Performance depends on the quality of the transition decisions.
- Complex relationships may be difficult to capture.

Graph-Based Parsing

Graph-based parsing treats dependency parsing as a graph optimization problem. Words are represented as nodes and possible dependency relationships are represented as edges. The system searches for the best overall dependency structure.

Advantages:

- Considers relationships globally.
- Can produce accurate dependency structures.
- Less affected by individual early decisions.
- Suitable when parsing accuracy is the main requirement.

Limitations:

- Can require greater computational resources.
- Usually more computationally expensive than transition-based parsing.
- Processing very large datasets can be slower.

Comparison and Conclusion

Transition-based parsing focuses on making sequential decisions and is generally faster. Graph-based parsing evaluates the overall dependency structure and can provide stronger global analysis but may require more computation. For large-scale applications where speed and scalability are important,

transition-based parsing is generally more suitable. However, graph-based parsing can be preferred when higher parsing accuracy is more important than processing speed.

Overall Conclusion

The choice of parsing technique depends on the application requirements. CFG is useful for hierarchical grammatical representation, dependency parsing is effective for word relationships, Earley parsing handles dynamic and ambiguous input, PCFG and neural parsing improve ambiguity resolution, feature structures support grammatical constraints, and transition-based parsing provides efficient dependency parsing for large-scale systems. These techniques can also be combined to achieve better accuracy, efficiency, and scalability in practical NLP applications.

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation

Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand
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Q. No. / Criteria Level	Scope of Items	Comparison Depth	Use of Evidence	Critical Thinking	Clarity of Presentation	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____